

PrimeCore Minerals

P5.4 Test network performance under heavy traffic

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Introduction

A reliable network must be able to withstand being under load when ports are maxed out at their top speed and not only being able to stay on but also process all the packets accordingly. This was determined early in the project and is number P5.4 on our scope.

The three main tests to confirm the reliability of a network include latency, bandwidth speed and jitter.

Latency confirms how long it takes network packets to travel between the 2 network devices with the lower the number the better. It is measured in milliseconds.

Bandwidth demonstrates the maximum speed a port can maintain.

Jitter shows the time variation between packets arriving. It is also known as network shaking and while some is expected, a large jitter shows congestion and packets arriving out of order, which will take more time for the switch to organise and therefore putting the switch under load.

A reliable network will allow the security/penetration tests to be conducted successfully as the network reliability will not affect the results.

Purpose

To test this scenario, a port limit of 100Mbps for standard ports and 1Gbps for trunk ports was placed on the LAN switches. This limit is lower than the switches recommended actual speed but is placed at this to protect the virtual environment from crashing.

Ping is the command that will be used to show latency between switches and is built into linux.

To demonstrate the analogy of the ports under maximum load, iperf3 has been installed on the switches and APs on the site side to simulate the load.

Tests have been completed from each switch to demonstrate that each switch is able to withstand the ability of being under load.

Iperf3 command and switches used in testing

The commands used to run the tests are as follows.

1. Receiving switch
 - a. `iperf3 -s`

2. Sending switch

- a. To run maximum speed test with TCP packets
 - i. Iperf3 -t 40 -c <receiving switch IP address>
- b. To test for jitter using UDP packets
 - i. Iperf3 -u -b 0 -l 1400 -t 40 -c <receiving switch IP address>

The iperf3 online documentation can be found here <https://iperf.fr/iperf-doc.php> with a list of the switches used below

-s = server (in this project switch/AP) receiving the results and replying

-t = time taken to run the test shown in seconds

-c = server (in this project switch/AP) sending the results and controlling the test

-u = use UDP packets only. This is to flood the network

-b 0 = bandwidth of 0 means flood the network as much as you can

-l 1400 = limit packet size to 1400 bytes. Network standard is 1500 bytes so this is below that to prevent packet fragmentation

Switch Speed Test Results

The following table is an overview of the results from the tests

Switch/AP Name	Ping/Speed/Jitter Test	Results
PCM-SITEA-SW01	Ping Test	0.680ms 0 packet loss
PCM-SITEA-SW01	Speed Test	796Mbps
PCM-SITEA-SW01	Jitter Test	0.059ms
PCM-SITEA-SW02	Ping Test	0.566ms 0 packet loss
PCM-SITEA-SW02	Speed Test	790Mbps
PCM-SITEA-SW02	Jitter Test	0.042ms
PCM-SITEB-SW01	Ping Test	0.686ms 0 packet loss
PCM-SITEB-SW01	Speed Test	860Mbps
PCM-SITEB-SW01	Jitter Test	0.45ms
PCM-SITEC-SW01	Ping Test	0.827ms 0 packet loss
PCM-SITEC-SW01	Speed Test	860Mbps
PCM-SITEC-SW01	Jitter Test	0.056ms
PCM-SITED-SW01	Ping Test	0.661ms

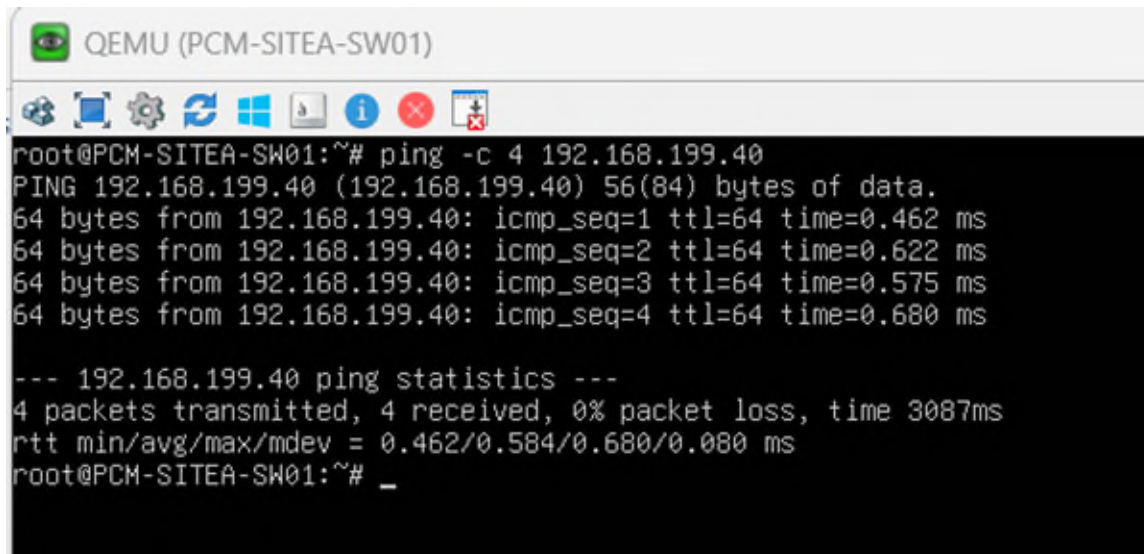
		0 packet loss
PCM-SITED-SW01	Speed Test	860Mbps
PCM-SITED-SW01	Jitter Test	0.048ms
PCM-SITEA-AP01	Ping Test	0.727ms 0 packet loss
PCM-SITEA-AP01	Speed Test	861Mbps
PCM-SITEA-AP01	Jitter Test	0.045ms
PCM-SITEA-AP02	Ping Test	0.624ms 0 packet loss
PCM-SITEA-AP02	Speed Test	856Mbps
PCM-SITEA-AP02	Jitter Test	0.045ms

Switch Results

Below are the screenshots taken from each switch/AP test

PCM-SITEA-SW01

PCM-SITEA-SW01 Ping Test



```

QEMU (PCM-SITEA-SW01)
root@PCM-SITEA-SW01:~# ping -c 4 192.168.199.40
PING 192.168.199.40 (192.168.199.40) 56(84) bytes of data.
64 bytes from 192.168.199.40: icmp_seq=1 ttl=64 time=0.462 ms
64 bytes from 192.168.199.40: icmp_seq=2 ttl=64 time=0.622 ms
64 bytes from 192.168.199.40: icmp_seq=3 ttl=64 time=0.575 ms
64 bytes from 192.168.199.40: icmp_seq=4 ttl=64 time=0.680 ms

--- 192.168.199.40 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.462/0.584/0.680/0.080 ms
root@PCM-SITEA-SW01:~# _

```

Figure 1 – screenshot of PCM-SITEA-SW01 ping test

PCM-SITEA-SW01 Speed Test

```

QEMU (PCM-SITEA-SW01)
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 5]  0.00-1.00    sec  97.6 MBytes  818 Mbits/sec    0  1.23 MBytes
[ 5]  1.00-2.00    sec  95.8 MBytes  803 Mbits/sec    0  1.62 MBytes
[ 5]  2.00-3.00    sec  92.4 MBytes  775 Mbits/sec    0  2.33 MBytes
[ 5]  3.00-4.00    sec  98.1 MBytes  822 Mbits/sec    0  2.33 MBytes
[ 5]  4.00-5.00    sec  96.4 MBytes  809 Mbits/sec    0  2.33 MBytes
[ 5]  5.00-6.00    sec  96.8 MBytes  811 Mbits/sec    0  2.33 MBytes
[ 5]  6.00-7.00    sec  95.4 MBytes  800 Mbits/sec    0  2.46 MBytes
[ 5]  7.00-8.00    sec  94.9 MBytes  796 Mbits/sec    0  2.46 MBytes
[ 5]  8.00-9.00    sec  97.6 MBytes  819 Mbits/sec    0  2.46 MBytes
[ 5]  9.00-10.00   sec  95.9 MBytes  805 Mbits/sec    0  2.46 MBytes
[ 5] 10.00-11.00   sec  96.2 MBytes  807 Mbits/sec    0  2.46 MBytes
[ 5] 11.00-12.00   sec  96.6 MBytes  812 Mbits/sec    0  2.46 MBytes
[ 5] 12.00-13.00   sec  96.5 MBytes  810 Mbits/sec    0  2.87 MBytes
[ 5] 13.00-14.00   sec  97.8 MBytes  819 Mbits/sec    0  2.87 MBytes
[ 5] 14.00-15.00   sec  95.6 MBytes  803 Mbits/sec    0  2.87 MBytes
[ 5] 15.00-16.00   sec  96.6 MBytes  810 Mbits/sec    0  2.87 MBytes
[ 5] 16.00-17.00   sec  95.1 MBytes  798 Mbits/sec    0  2.87 MBytes
[ 5] 17.00-18.00   sec  91.6 MBytes  768 Mbits/sec    0  3.09 MBytes
[ 5] 18.00-19.00   sec  95.6 MBytes  802 Mbits/sec    0  3.09 MBytes
[ 5] 19.00-20.00   sec  88.8 MBytes  745 Mbits/sec    0  4.04 MBytes
[ 5] 20.00-21.00   sec  92.5 MBytes  776 Mbits/sec    0  4.04 MBytes
[ 5] 21.00-22.00   sec  90.1 MBytes  756 Mbits/sec    0  4.04 MBytes
[ 5] 22.00-23.00   sec  94.0 MBytes  788 Mbits/sec    0  4.04 MBytes
[ 5] 23.00-24.00   sec  94.9 MBytes  796 Mbits/sec    0  4.04 MBytes
[ 5] 24.00-25.00   sec  95.8 MBytes  803 Mbits/sec    0  4.04 MBytes
[ 5] 25.00-26.00   sec  95.6 MBytes  802 Mbits/sec    0  4.04 MBytes
[ 5] 26.00-27.00   sec  96.4 MBytes  808 Mbits/sec    0  4.04 MBytes
[ 5] 27.00-28.00   sec  94.4 MBytes  792 Mbits/sec    0  4.04 MBytes
[ 5] 28.00-29.00   sec  96.2 MBytes  807 Mbits/sec    0  4.04 MBytes
[ 5] 29.00-30.00   sec  96.9 MBytes  812 Mbits/sec    0  4.04 MBytes
[ 5] 30.00-31.00   sec  93.4 MBytes  783 Mbits/sec    0  4.04 MBytes
[ 5] 31.00-32.00   sec  94.5 MBytes  793 Mbits/sec    0  4.04 MBytes
[ 5] 32.00-33.00   sec  90.0 MBytes  755 Mbits/sec    0  4.04 MBytes
[ 5] 33.00-34.00   sec  95.0 MBytes  797 Mbits/sec    0  4.04 MBytes
[ 5] 34.00-35.00   sec  94.9 MBytes  796 Mbits/sec    0  4.04 MBytes
[ 5] 35.00-36.00   sec  90.4 MBytes  758 Mbits/sec    0  4.04 MBytes
[ 5] 36.00-37.00   sec  91.5 MBytes  768 Mbits/sec    0  4.04 MBytes
[ 5] 37.00-38.00   sec  97.0 MBytes  814 Mbits/sec    0  4.04 MBytes
[ 5] 38.00-39.00   sec  96.4 MBytes  808 Mbits/sec    0  4.04 MBytes
[ 5] 39.00-40.00   sec  97.8 MBytes  820 Mbits/sec    0  4.04 MBytes
-- -- -- -- --
[ ID] Interval      Transfer      Bitrate      Retr
[ 5]  0.00-40.00   sec  3.71 GBytes  797 Mbits/sec    0      sender
[ 5]  0.00-40.02   sec  3.71 GBytes  796 Mbits/sec              receiver

iperf Done.
root@PCM-SITEA-SW01:~#

```

Figure 2 – screenshot of PCM-SITEA-SW01 speed test

PCM-SITEA-SW01 Jitter Test

```

QEMU (PCM-SITEA-SW01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5]  0.00-1.00    sec  47.5 MBytes  398 Mbits/sec  35577
[ 5]  1.00-2.00    sec  45.2 MBytes  379 Mbits/sec  33836
[ 5]  2.00-3.00    sec  46.9 MBytes  393 Mbits/sec  35137
[ 5]  3.00-4.05    sec  44.2 MBytes  356 Mbits/sec  33132
[ 5]  4.05-5.00    sec  44.8 MBytes  394 Mbits/sec  33584
[ 5]  5.00-6.00    sec  43.0 MBytes  361 Mbits/sec  32216
[ 5]  6.00-7.00    sec  42.6 MBytes  358 Mbits/sec  31922
[ 5]  7.00-8.00    sec  45.2 MBytes  379 Mbits/sec  33848
[ 5]  8.00-9.00    sec  44.2 MBytes  371 Mbits/sec  33140
[ 5]  9.00-10.00   sec  40.4 MBytes  339 Mbits/sec  30228
[ 5] 10.00-11.00   sec  48.4 MBytes  406 Mbits/sec  36269
[ 5] 11.00-12.00   sec  43.1 MBytes  361 Mbits/sec  32305
[ 5] 12.00-13.00   sec  47.6 MBytes  400 Mbits/sec  35664
[ 5] 13.00-14.00   sec  47.0 MBytes  394 Mbits/sec  35174
[ 5] 14.00-15.00   sec  44.8 MBytes  375 Mbits/sec  33518
[ 5] 15.00-16.00   sec  49.8 MBytes  418 Mbits/sec  37321
[ 5] 16.00-17.00   sec  45.2 MBytes  379 Mbits/sec  33831
[ 5] 17.00-18.00   sec  44.6 MBytes  375 Mbits/sec  33440
[ 5] 18.00-19.00   sec  43.6 MBytes  366 Mbits/sec  32655
[ 5] 19.00-20.00   sec  45.1 MBytes  378 Mbits/sec  33752
[ 5] 20.00-21.00   sec  48.3 MBytes  405 Mbits/sec  36184
[ 5] 21.00-22.00   sec  44.1 MBytes  370 Mbits/sec  33038
[ 5] 22.00-23.00   sec  46.8 MBytes  393 Mbits/sec  35044
[ 5] 23.00-24.00   sec  49.8 MBytes  417 Mbits/sec  37290
[ 5] 24.00-25.00   sec  44.5 MBytes  374 Mbits/sec  33343
[ 5] 25.00-26.00   sec  51.1 MBytes  428 Mbits/sec  38276
[ 5] 26.00-27.00   sec  42.2 MBytes  354 Mbits/sec  31576
[ 5] 27.00-28.00   sec  42.5 MBytes  356 Mbits/sec  31815
[ 5] 28.00-29.00   sec  47.5 MBytes  398 Mbits/sec  35590
[ 5] 29.00-30.00   sec  41.6 MBytes  350 Mbits/sec  31187
[ 5] 30.00-31.00   sec  42.9 MBytes  360 Mbits/sec  32148
[ 5] 31.00-32.00   sec  45.2 MBytes  379 Mbits/sec  33859
[ 5] 32.00-33.00   sec  47.9 MBytes  402 Mbits/sec  35880
[ 5] 33.00-34.00   sec  49.6 MBytes  416 Mbits/sec  37118
[ 5] 34.00-35.00   sec  46.1 MBytes  387 Mbits/sec  34548
[ 5] 35.00-36.00   sec  47.4 MBytes  397 Mbits/sec  35472
[ 5] 36.00-37.00   sec  46.6 MBytes  391 Mbits/sec  34878
[ 5] 37.00-38.00   sec  49.4 MBytes  415 Mbits/sec  37026
[ 5] 38.00-39.00   sec  45.5 MBytes  382 Mbits/sec  34083
[ 5] 39.00-40.00   sec  50.7 MBytes  425 Mbits/sec  37996
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5]  0.00-40.00    sec  1.79 GBytes  384 Mbits/sec  0.000 ms    0/1372900 (0%) sender
[ 5]  0.00-40.06    sec  1.54 GBytes  331 Mbits/sec  0.059 ms   190043/1372900 (14%) receiver

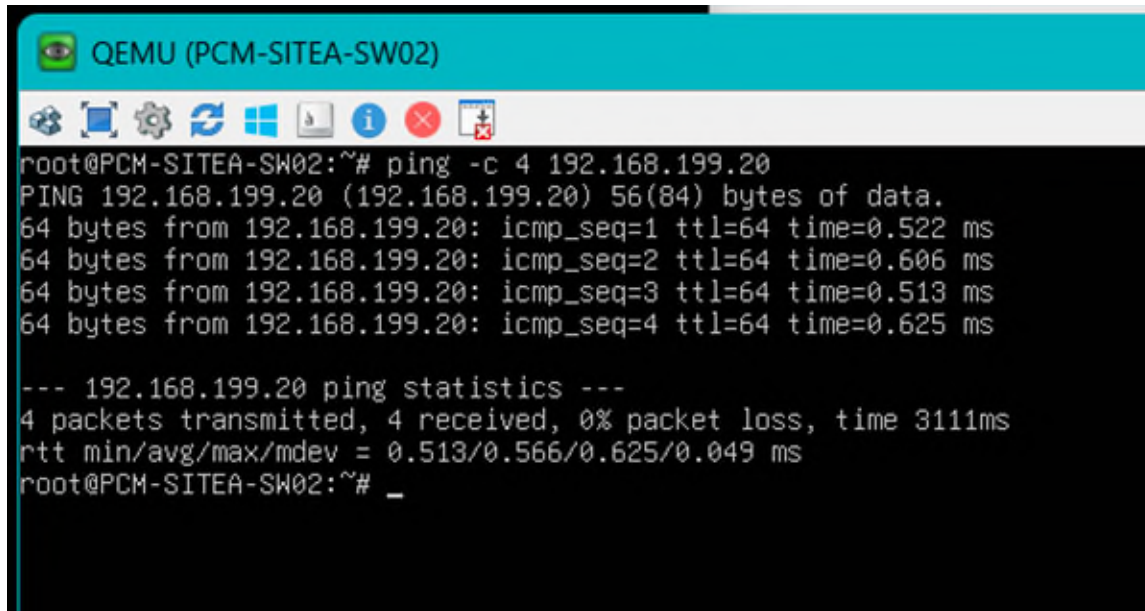
iperf Done.
root@PCM-SITEA-SW01:~#

```

Figure 3 – screenshot of PCM-SITEA-SW01 jitter test

PCM-SITEA-SW02

PCM-SITEA-SW02 Ping Test



```
QEMU (PCM-SITEA-SW02)
root@PCM-SITEA-SW02:~# ping -c 4 192.168.199.20
PING 192.168.199.20 (192.168.199.20) 56(84) bytes of data.
64 bytes from 192.168.199.20: icmp_seq=1 ttl=64 time=0.522 ms
64 bytes from 192.168.199.20: icmp_seq=2 ttl=64 time=0.606 ms
64 bytes from 192.168.199.20: icmp_seq=3 ttl=64 time=0.513 ms
64 bytes from 192.168.199.20: icmp_seq=4 ttl=64 time=0.625 ms

--- 192.168.199.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3111ms
rtt min/avg/max/mdev = 0.513/0.566/0.625/0.049 ms
root@PCM-SITEA-SW02:~# _
```

Figure 4 – screenshot of PCM-SITEA-SW02 ping test

PCM-SITEA-SW02 Speed Test

```

[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 5]  0.00-1.00    sec  83.5 MBytes  700 Mbits/sec    0  1.37 MBytes
[ 5]  1.00-2.00    sec  94.2 MBytes  791 Mbits/sec    0  1.71 MBytes
[ 5]  2.00-3.00    sec  95.9 MBytes  805 Mbits/sec    0  1.80 MBytes
[ 5]  3.00-4.00    sec  97.2 MBytes  816 Mbits/sec    0  1.80 MBytes
[ 5]  4.00-5.00    sec  79.0 MBytes  662 Mbits/sec    0  1.98 MBytes
[ 5]  5.00-6.00    sec  95.0 MBytes  798 Mbits/sec    0  1.98 MBytes
[ 5]  6.00-7.00    sec  96.0 MBytes  805 Mbits/sec    0  1.98 MBytes
[ 5]  7.00-8.00    sec  94.2 MBytes  790 Mbits/sec    0  2.08 MBytes
[ 5]  8.00-9.00    sec  89.4 MBytes  751 Mbits/sec    0  3.21 MBytes
[ 5]  9.00-10.00   sec  92.4 MBytes  775 Mbits/sec    0  3.21 MBytes
[ 5] 10.00-11.00   sec  97.9 MBytes  821 Mbits/sec    0  3.21 MBytes
[ 5] 11.00-12.00   sec  94.9 MBytes  796 Mbits/sec    0  3.21 MBytes
[ 5] 12.00-13.00   sec  94.9 MBytes  796 Mbits/sec    0  3.21 MBytes
[ 5] 13.00-14.00   sec  95.5 MBytes  800 Mbits/sec    0  3.21 MBytes
[ 5] 14.00-15.00   sec  96.5 MBytes  810 Mbits/sec    0  3.21 MBytes
[ 5] 15.00-16.00   sec  96.5 MBytes  810 Mbits/sec    0  3.21 MBytes
[ 5] 16.00-17.00   sec  96.9 MBytes  813 Mbits/sec    0  3.21 MBytes
[ 5] 17.00-18.00   sec  96.2 MBytes  807 Mbits/sec    0  3.21 MBytes
[ 5] 18.00-19.00   sec  95.4 MBytes  801 Mbits/sec    0  3.21 MBytes
[ 5] 19.00-20.00   sec  94.8 MBytes  794 Mbits/sec    0  3.21 MBytes
[ 5] 20.00-21.00   sec  94.1 MBytes  791 Mbits/sec    0  3.21 MBytes
[ 5] 21.00-22.00   sec  92.9 MBytes  778 Mbits/sec    0  3.21 MBytes
[ 5] 22.00-23.00   sec  93.4 MBytes  783 Mbits/sec    0  3.21 MBytes
[ 5] 23.00-24.00   sec  94.5 MBytes  793 Mbits/sec    0  3.21 MBytes
[ 5] 24.00-25.00   sec  95.4 MBytes  800 Mbits/sec    0  3.21 MBytes
[ 5] 25.00-26.00   sec  94.8 MBytes  795 Mbits/sec    0  4.89 MBytes
[ 5] 26.00-27.00   sec  98.5 MBytes  826 Mbits/sec    0  4.89 MBytes
[ 5] 27.00-28.00   sec  96.5 MBytes  809 Mbits/sec    0  4.89 MBytes
[ 5] 28.00-29.00   sec  95.0 MBytes  797 Mbits/sec    0  4.89 MBytes
[ 5] 29.00-30.00   sec  96.5 MBytes  810 Mbits/sec    0  4.89 MBytes
[ 5] 30.00-31.00   sec  96.5 MBytes  809 Mbits/sec    0  4.89 MBytes
[ 5] 31.00-32.00   sec  97.2 MBytes  815 Mbits/sec    0  4.89 MBytes
[ 5] 32.00-33.00   sec  95.1 MBytes  799 Mbits/sec    0  4.89 MBytes
[ 5] 33.00-34.00   sec  96.8 MBytes  811 Mbits/sec    0  4.89 MBytes
[ 5] 34.00-35.00   sec  95.5 MBytes  802 Mbits/sec    0  4.89 MBytes
[ 5] 35.00-36.00   sec  93.6 MBytes  785 Mbits/sec    0  4.89 MBytes
[ 5] 36.00-37.00   sec  95.1 MBytes  798 Mbits/sec    0  4.89 MBytes
[ 5] 37.00-38.00   sec  92.5 MBytes  776 Mbits/sec    0  4.89 MBytes
[ 5] 38.00-39.00   sec  91.8 MBytes  769 Mbits/sec    0  4.89 MBytes
[ 5] 39.00-40.00   sec  86.9 MBytes  729 Mbits/sec    0  4.89 MBytes
- - - - -
[ ID] Interval      Transfer      Bitrate      Retr
[ 5]  0.00-40.00   sec  3.68 GBytes  790 Mbits/sec    0
[ 5]  0.00-40.03   sec  3.68 GBytes  790 Mbits/sec    0
sender
receiver

iperf Done.
root@PCM-SITEA-SW02:~#

```

Figure 5 – screenshot of PCM-SITEA-SW02 speed test

PCM-SITEA-SW02 Jitter Test

```

QEMU (PCM-SITEA-SW02)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00      sec 47.3 MBytes 396 Mbits/sec 35419
[ 5] 1.00-2.00      sec 46.5 MBytes 391 Mbits/sec 34856
[ 5] 2.00-3.00      sec 46.8 MBytes 392 Mbits/sec 35053
[ 5] 3.00-4.00      sec 44.7 MBytes 375 Mbits/sec 33470
[ 5] 4.00-5.00      sec 48.3 MBytes 405 Mbits/sec 36154
[ 5] 5.00-6.00      sec 47.2 MBytes 396 Mbits/sec 35340
[ 5] 6.00-7.00      sec 42.8 MBytes 359 Mbits/sec 32069
[ 5] 7.00-8.00      sec 45.6 MBytes 383 Mbits/sec 34127
[ 5] 8.00-9.00      sec 37.2 MBytes 312 Mbits/sec 27884
[ 5] 9.00-10.00     sec 46.8 MBytes 393 Mbits/sec 35060
[ 5] 10.00-11.00    sec 47.7 MBytes 400 Mbits/sec 35695
[ 5] 11.00-12.00    sec 45.9 MBytes 385 Mbits/sec 34402
[ 5] 12.00-13.00    sec 45.0 MBytes 378 Mbits/sec 33717
[ 5] 13.00-14.00    sec 45.9 MBytes 385 Mbits/sec 34409
[ 5] 14.00-15.00    sec 41.1 MBytes 345 Mbits/sec 30761
[ 5] 15.00-16.00    sec 46.8 MBytes 392 Mbits/sec 35035
[ 5] 16.00-17.00    sec 44.6 MBytes 374 Mbits/sec 33418
[ 5] 17.00-18.00    sec 47.1 MBytes 395 Mbits/sec 35284
[ 5] 18.00-19.00    sec 45.2 MBytes 379 Mbits/sec 33862
[ 5] 19.00-20.00    sec 44.8 MBytes 375 Mbits/sec 33558
[ 5] 20.00-21.00    sec 48.2 MBytes 404 Mbits/sec 36076
[ 5] 21.00-22.00    sec 45.1 MBytes 379 Mbits/sec 33796
[ 5] 22.00-23.00    sec 46.6 MBytes 390 Mbits/sec 34898
[ 5] 23.00-24.00    sec 43.8 MBytes 368 Mbits/sec 32801
[ 5] 24.00-25.00    sec 46.5 MBytes 389 Mbits/sec 34801
[ 5] 25.00-26.00    sec 45.8 MBytes 384 Mbits/sec 34323
[ 5] 26.00-27.00    sec 43.5 MBytes 365 Mbits/sec 32573
[ 5] 27.00-28.00    sec 48.3 MBytes 406 Mbits/sec 36211
[ 5] 28.00-29.00    sec 47.1 MBytes 395 Mbits/sec 35273
[ 5] 29.00-30.00    sec 47.6 MBytes 399 Mbits/sec 35620
[ 5] 30.00-31.00    sec 43.3 MBytes 363 Mbits/sec 32411
[ 5] 31.00-32.00    sec 42.9 MBytes 361 Mbits/sec 32121
[ 5] 32.00-33.00    sec 47.6 MBytes 399 Mbits/sec 35638
[ 5] 33.00-34.00    sec 44.6 MBytes 374 Mbits/sec 33397
[ 5] 34.00-35.00    sec 45.4 MBytes 380 Mbits/sec 33973
[ 5] 35.00-36.00    sec 45.2 MBytes 379 Mbits/sec 33844
[ 5] 36.00-37.00    sec 45.8 MBytes 384 Mbits/sec 34306
[ 5] 37.00-38.00    sec 43.4 MBytes 363 Mbits/sec 32472
[ 5] 38.00-39.00    sec 47.4 MBytes 398 Mbits/sec 35497
[ 5] 39.00-40.00    sec 52.6 MBytes 441 Mbits/sec 39416
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-40.00     sec 1.78 GBytes 383 Mbits/sec 0.000 ms    0/1369020 (0%) sender
[ 5] 0.00-40.07     sec 1.58 GBytes 338 Mbits/sec 0.042 ms    159242/1369020 (12%) receiver

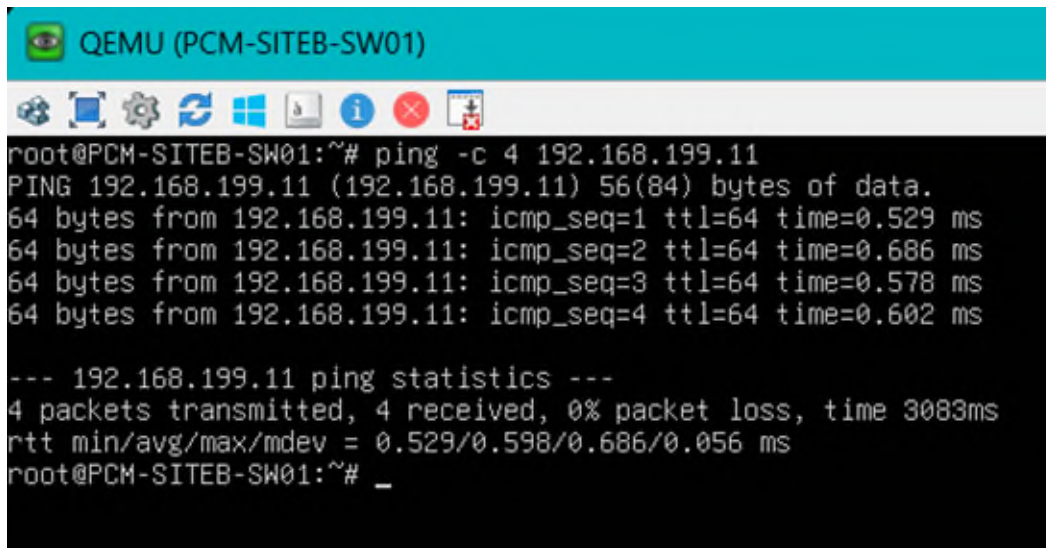
iperf Done.
root@PCM-SITEA-SW02:~#

```

Figure 6 – screenshot of PCM-SITEA-SW02 jitter test

PCM-SITEB-SW01

PCM-SITEB-SW01 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITEB-SW01)". The terminal displays the output of a ping command executed from the root user at the PCM-SITEB-SW01 host. The command is "ping -c 4 192.168.199.11". The output shows four successful ping requests, each with a 64-byte payload, an ICMP sequence number, a TTL of 64, and a response time. The response times are 0.529 ms, 0.686 ms, 0.578 ms, and 0.602 ms. Below the individual results, a summary line shows "4 packets transmitted, 4 received, 0% packet loss, time 3083ms". The final line shows the round-trip time (rtt) statistics: "rtt min/avg/max/mdev = 0.529/0.598/0.686/0.056 ms". The terminal prompt returns to "root@PCM-SITEB-SW01:~#".

```
root@PCM-SITEB-SW01:~# ping -c 4 192.168.199.11
PING 192.168.199.11 (192.168.199.11) 56(84) bytes of data.
64 bytes from 192.168.199.11: icmp_seq=1 ttl=64 time=0.529 ms
64 bytes from 192.168.199.11: icmp_seq=2 ttl=64 time=0.686 ms
64 bytes from 192.168.199.11: icmp_seq=3 ttl=64 time=0.578 ms
64 bytes from 192.168.199.11: icmp_seq=4 ttl=64 time=0.602 ms

--- 192.168.199.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3083ms
rtt min/avg/max/mdev = 0.529/0.598/0.686/0.056 ms
root@PCM-SITEB-SW01:~# _
```

Figure 7 – screenshot of PCM-SITEB-SW01 ping test

PCM-SITEB-SW01 Speed Test

QEMU (PCM-SITEB-SW01)

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	103 MBytes	866 Mbits/sec	0	689 KBytes
[5]	1.00-2.00	sec	101 MBytes	845 Mbits/sec	0	689 KBytes
[5]	2.00-3.00	sec	106 MBytes	886 Mbits/sec	0	689 KBytes
[5]	3.00-4.00	sec	101 MBytes	846 Mbits/sec	0	725 KBytes
[5]	4.00-5.00	sec	105 MBytes	879 Mbits/sec	0	725 KBytes
[5]	5.00-6.00	sec	98.8 MBytes	828 Mbits/sec	0	765 KBytes
[5]	6.00-7.00	sec	104 MBytes	876 Mbits/sec	0	765 KBytes
[5]	7.00-8.00	sec	105 MBytes	878 Mbits/sec	0	765 KBytes
[5]	8.00-9.00	sec	102 MBytes	852 Mbits/sec	0	803 KBytes
[5]	9.00-10.00	sec	103 MBytes	866 Mbits/sec	0	803 KBytes
[5]	10.00-11.00	sec	102 MBytes	854 Mbits/sec	0	884 KBytes
[5]	11.00-12.00	sec	103 MBytes	864 Mbits/sec	0	884 KBytes
[5]	12.00-13.00	sec	103 MBytes	866 Mbits/sec	0	884 KBytes
[5]	13.00-14.00	sec	101 MBytes	846 Mbits/sec	0	1.57 MBytes
[5]	14.00-15.00	sec	104 MBytes	874 Mbits/sec	0	1.57 MBytes
[5]	15.00-16.00	sec	101 MBytes	850 Mbits/sec	0	1.57 MBytes
[5]	16.00-17.00	sec	104 MBytes	872 Mbits/sec	0	1.57 MBytes
[5]	17.00-18.00	sec	103 MBytes	864 Mbits/sec	0	1.57 MBytes
[5]	18.00-19.00	sec	101 MBytes	845 Mbits/sec	0	2.47 MBytes
[5]	19.00-20.00	sec	102 MBytes	852 Mbits/sec	0	2.47 MBytes
[5]	20.00-21.00	sec	97.6 MBytes	818 Mbits/sec	0	2.47 MBytes
[5]	21.00-22.00	sec	103 MBytes	864 Mbits/sec	0	2.47 MBytes
[5]	22.00-23.00	sec	102 MBytes	854 Mbits/sec	0	2.47 MBytes
[5]	23.00-24.00	sec	100 MBytes	842 Mbits/sec	0	2.47 MBytes
[5]	24.00-25.00	sec	103 MBytes	867 Mbits/sec	0	2.47 MBytes
[5]	25.00-26.00	sec	102 MBytes	857 Mbits/sec	0	2.47 MBytes
[5]	26.00-27.00	sec	103 MBytes	864 Mbits/sec	0	2.47 MBytes
[5]	27.00-28.00	sec	105 MBytes	878 Mbits/sec	0	2.47 MBytes
[5]	28.00-29.00	sec	102 MBytes	854 Mbits/sec	0	2.47 MBytes
[5]	29.00-30.00	sec	104 MBytes	877 Mbits/sec	0	2.47 MBytes
[5]	30.00-31.00	sec	104 MBytes	872 Mbits/sec	0	3.78 MBytes
[5]	31.00-32.00	sec	103 MBytes	864 Mbits/sec	0	3.78 MBytes
[5]	32.00-33.00	sec	102 MBytes	855 Mbits/sec	0	3.78 MBytes
[5]	33.00-34.00	sec	98.9 MBytes	827 Mbits/sec	0	3.78 MBytes
[5]	34.00-35.00	sec	102 MBytes	859 Mbits/sec	0	3.78 MBytes
[5]	35.00-36.01	sec	103 MBytes	853 Mbits/sec	0	3.78 MBytes
[5]	36.01-37.00	sec	102 MBytes	865 Mbits/sec	0	3.78 MBytes
[5]	37.00-38.00	sec	105 MBytes	878 Mbits/sec	0	3.78 MBytes
[5]	38.00-39.00	sec	102 MBytes	852 Mbits/sec	0	3.78 MBytes
[5]	39.00-40.00	sec	104 MBytes	875 Mbits/sec	0	3.78 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	4.01 GBytes	860 Mbits/sec	0	sender
[5]	0.00-40.01	sec	4.00 GBytes	860 Mbits/sec		receiver

iperf Done.
root@PCM-SITEB-SW01:~#

Figure 8 – screenshot of PCM-SITEB-SW01 speed test

PCM-SITEB-SW01 Jitter Test

```

QEMU (PCM-SITEB-SW01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5]  0.00-1.00    sec  46.2 MBytes  388 Mbits/sec  34623
[ 5]  1.00-2.00    sec  47.0 MBytes  394 Mbits/sec  35210
[ 5]  2.00-3.00    sec  45.6 MBytes  382 Mbits/sec  34123
[ 5]  3.00-4.00    sec  42.4 MBytes  355 Mbits/sec  31778
[ 5]  4.00-5.00    sec  46.1 MBytes  387 Mbits/sec  34493
[ 5]  5.00-6.00    sec  44.3 MBytes  372 Mbits/sec  33170
[ 5]  6.00-7.00    sec  45.6 MBytes  382 Mbits/sec  34132
[ 5]  7.00-8.00    sec  45.7 MBytes  384 Mbits/sec  34245
[ 5]  8.00-9.00    sec  45.3 MBytes  380 Mbits/sec  33917
[ 5]  9.00-10.00   sec  43.7 MBytes  367 Mbits/sec  32745
[ 5] 10.00-11.00   sec  43.6 MBytes  366 Mbits/sec  32652
[ 5] 11.00-12.00   sec  45.6 MBytes  383 Mbits/sec  34161
[ 5] 12.00-13.00   sec  44.6 MBytes  374 Mbits/sec  33421
[ 5] 13.00-14.00   sec  47.1 MBytes  395 Mbits/sec  35257
[ 5] 14.00-15.00   sec  47.9 MBytes  402 Mbits/sec  35884
[ 5] 15.00-16.00   sec  47.6 MBytes  398 Mbits/sec  35623
[ 5] 16.00-17.00   sec  44.7 MBytes  375 Mbits/sec  33449
[ 5] 17.00-18.00   sec  42.6 MBytes  357 Mbits/sec  31919
[ 5] 18.00-19.00   sec  44.1 MBytes  370 Mbits/sec  33044
[ 5] 19.00-20.00   sec  44.9 MBytes  376 Mbits/sec  33612
[ 5] 20.00-21.00   sec  47.6 MBytes  399 Mbits/sec  35665
[ 5] 21.00-22.00   sec  46.9 MBytes  393 Mbits/sec  35094
[ 5] 22.00-23.00   sec  44.1 MBytes  370 Mbits/sec  33000
[ 5] 23.00-24.00   sec  45.1 MBytes  378 Mbits/sec  33780
[ 5] 24.00-25.00   sec  44.3 MBytes  372 Mbits/sec  33180
[ 5] 25.00-26.00   sec  44.9 MBytes  377 Mbits/sec  33639
[ 5] 26.00-27.00   sec  45.5 MBytes  382 Mbits/sec  34065
[ 5] 27.00-28.00   sec  45.1 MBytes  378 Mbits/sec  33747
[ 5] 28.00-29.00   sec  44.8 MBytes  376 Mbits/sec  33534
[ 5] 29.00-30.00   sec  45.8 MBytes  384 Mbits/sec  34289
[ 5] 30.00-31.00   sec  44.5 MBytes  373 Mbits/sec  33310
[ 5] 31.00-32.00   sec  44.9 MBytes  376 Mbits/sec  33596
[ 5] 32.00-33.00   sec  45.1 MBytes  379 Mbits/sec  33804
[ 5] 33.00-34.00   sec  45.3 MBytes  380 Mbits/sec  33950
[ 5] 34.00-35.00   sec  46.4 MBytes  390 Mbits/sec  34790
[ 5] 35.00-36.00   sec  45.3 MBytes  380 Mbits/sec  33910
[ 5] 36.00-37.00   sec  43.3 MBytes  363 Mbits/sec  32454
[ 5] 37.00-38.00   sec  45.1 MBytes  378 Mbits/sec  33766
[ 5] 38.00-39.00   sec  46.0 MBytes  386 Mbits/sec  34423
[ 5] 39.00-40.00   sec  44.8 MBytes  375 Mbits/sec  33541
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5]  0.00-40.00    sec  1.77 GBytes  379 Mbits/sec  0.000 ms    0/1354995 (0%)  sender
[ 5]  0.00-40.00    sec  1.76 GBytes  378 Mbits/sec  0.045 ms    5747/1354995 (0.42%) receiver

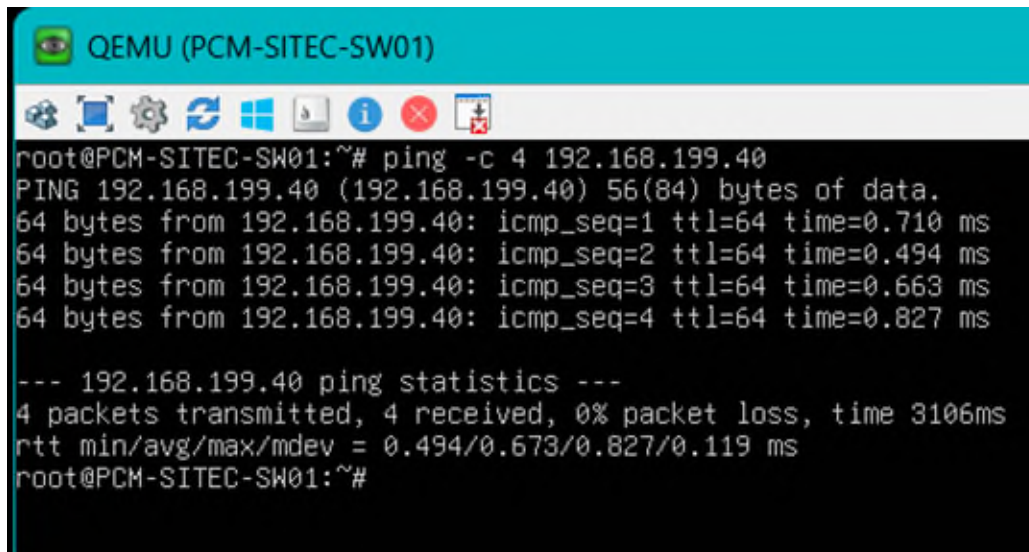
iperf Done.
root@PCM-SITEB-SW01:~#

```

Figure 9 – screenshot of PCM-SITEB-SW01 jitter test

PCM-SITEC-SW01

PCM-SITEC-SW01 Ping Test



```
root@PCM-SITEC-SW01:~# ping -c 4 192.168.199.40
PING 192.168.199.40 (192.168.199.40) 56(84) bytes of data.
64 bytes from 192.168.199.40: icmp_seq=1 ttl=64 time=0.710 ms
64 bytes from 192.168.199.40: icmp_seq=2 ttl=64 time=0.494 ms
64 bytes from 192.168.199.40: icmp_seq=3 ttl=64 time=0.663 ms
64 bytes from 192.168.199.40: icmp_seq=4 ttl=64 time=0.827 ms

--- 192.168.199.40 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3106ms
rtt min/avg/max/mdev = 0.494/0.673/0.827/0.119 ms
root@PCM-SITEC-SW01:~#
```

Figure 10 – screenshot of PCM-SITEC-SW01 ping test

PCM-SITEC-SW01 Speed Test

```

QEMU (PCM-SITEC-SW01)
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 5]  0.00-1.00    sec    103 MBytes    864 Mbits/sec    0    471 KBytes
[ 5]  1.00-2.00    sec    106 MBytes    886 Mbits/sec    0    494 KBytes
[ 5]  2.00-3.00    sec    106 MBytes    887 Mbits/sec    0    515 KBytes
[ 5]  3.00-4.00    sec    102 MBytes    854 Mbits/sec    0    568 KBytes
[ 5]  4.00-5.00    sec    102 MBytes    854 Mbits/sec    0    631 KBytes
[ 5]  5.00-6.00    sec    99.6 MBytes    836 Mbits/sec    0    701 KBytes
[ 5]  6.00-7.00    sec    105 MBytes    879 Mbits/sec    0    701 KBytes
[ 5]  7.00-8.00    sec    103 MBytes    863 Mbits/sec    0    701 KBytes
[ 5]  8.00-9.00    sec    98.9 MBytes    829 Mbits/sec    0    701 KBytes
[ 5]  9.00-10.00   sec    101 MBytes    844 Mbits/sec    0    773 KBytes
[ 5] 10.00-11.00   sec    102 MBytes    855 Mbits/sec    0    831 KBytes
[ 5] 11.00-12.00   sec    100 MBytes    841 Mbits/sec    0    831 KBytes
[ 5] 12.00-13.00   sec    103 MBytes    864 Mbits/sec    0    831 KBytes
[ 5] 13.00-14.00   sec    102 MBytes    852 Mbits/sec    0    1.23 MBytes
[ 5] 14.00-15.00   sec    102 MBytes    852 Mbits/sec    0    1.23 MBytes
[ 5] 15.00-16.00   sec    100 MBytes    839 Mbits/sec    0    1.87 MBytes
[ 5] 16.00-17.00   sec    104 MBytes    873 Mbits/sec    0    1.87 MBytes
[ 5] 17.00-18.00   sec    103 MBytes    862 Mbits/sec    0    1.87 MBytes
[ 5] 18.00-19.00   sec    101 MBytes    849 Mbits/sec    0    1.87 MBytes
[ 5] 19.00-20.00   sec    104 MBytes    873 Mbits/sec    0    1.87 MBytes
[ 5] 20.00-21.00   sec    101 MBytes    850 Mbits/sec    0    1.87 MBytes
[ 5] 21.00-22.00   sec    104 MBytes    875 Mbits/sec    0    1.87 MBytes
[ 5] 22.00-23.00   sec    104 MBytes    872 Mbits/sec    0    1.87 MBytes
[ 5] 23.00-24.00   sec    100 MBytes    841 Mbits/sec    0    1.87 MBytes
[ 5] 24.00-25.00   sec    105 MBytes    884 Mbits/sec    0    1.87 MBytes
[ 5] 25.00-26.00   sec    103 MBytes    862 Mbits/sec    0    1.87 MBytes
[ 5] 26.00-27.00   sec    104 MBytes    873 Mbits/sec    0    1.87 MBytes
[ 5] 27.00-28.00   sec    102 MBytes    854 Mbits/sec    0    1.87 MBytes
[ 5] 28.00-29.00   sec    104 MBytes    869 Mbits/sec    0    2.84 MBytes
[ 5] 29.00-30.00   sec    103 MBytes    862 Mbits/sec    0    2.84 MBytes
[ 5] 30.00-31.00   sec    104 MBytes    874 Mbits/sec    0    2.84 MBytes
[ 5] 31.00-32.00   sec    103 MBytes    861 Mbits/sec    0    2.84 MBytes
[ 5] 32.00-33.00   sec    104 MBytes    868 Mbits/sec    0    2.84 MBytes
[ 5] 33.00-34.00   sec    102 MBytes    856 Mbits/sec    0    2.84 MBytes
[ 5] 34.00-35.00   sec    103 MBytes    865 Mbits/sec    0    2.84 MBytes
[ 5] 35.00-36.00   sec    104 MBytes    874 Mbits/sec    0    2.84 MBytes
[ 5] 36.00-37.00   sec    103 MBytes    864 Mbits/sec    0    2.84 MBytes
[ 5] 37.00-38.00   sec    103 MBytes    863 Mbits/sec    0    2.84 MBytes
[ 5] 38.00-39.00   sec    102 MBytes    853 Mbits/sec    0    2.84 MBytes
[ 5] 39.00-40.00   sec    99.5 MBytes    835 Mbits/sec    0    2.84 MBytes
- - - - -
[ ID] Interval      Transfer      Bitrate      Retr
[ 5]  0.00-40.00   sec    4.01 GBytes    860 Mbits/sec    0
[ 5]  0.00-40.01   sec    4.00 GBytes    860 Mbits/sec    0
sender
receiver

iperf Done.
root@PCM-SITEC-SW01:~#

```

Figure 11 – screenshot of PCM-SITEC-SW01 speed test

PCM-SITEC-SW01 Jitter Test

```

QEMU (PCM-SITEC-SW01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00    sec 45.9 MBytes  385 Mbits/sec 34391
[ 5] 1.00-2.00    sec 47.9 MBytes  402 Mbits/sec 35877
[ 5] 2.00-3.00    sec 44.3 MBytes  372 Mbits/sec 33186
[ 5] 3.00-4.00    sec 47.1 MBytes  395 Mbits/sec 35296
[ 5] 4.00-5.00    sec 45.8 MBytes  384 Mbits/sec 34335
[ 5] 5.00-6.00    sec 45.4 MBytes  381 Mbits/sec 34021
[ 5] 6.00-7.00    sec 44.9 MBytes  377 Mbits/sec 33649
[ 5] 7.00-8.00    sec 43.9 MBytes  368 Mbits/sec 32892
[ 5] 8.00-9.00    sec 43.8 MBytes  367 Mbits/sec 32806
[ 5] 9.00-10.00   sec 43.2 MBytes  362 Mbits/sec 32319
[ 5] 10.00-11.00   sec 41.7 MBytes  350 Mbits/sec 31230
[ 5] 11.00-12.00   sec 43.4 MBytes  364 Mbits/sec 32536
[ 5] 12.00-13.00   sec 41.6 MBytes  349 Mbits/sec 31155
[ 5] 13.00-14.00   sec 43.1 MBytes  362 Mbits/sec 32292
[ 5] 14.00-15.00   sec 46.0 MBytes  386 Mbits/sec 34454
[ 5] 15.00-16.00   sec 47.2 MBytes  396 Mbits/sec 35369
[ 5] 16.00-17.00   sec 45.3 MBytes  380 Mbits/sec 33899
[ 5] 17.00-18.00   sec 45.4 MBytes  381 Mbits/sec 33983
[ 5] 18.00-19.00   sec 42.6 MBytes  358 Mbits/sec 31938
[ 5] 19.00-20.00   sec 47.7 MBytes  401 Mbits/sec 35749
[ 5] 20.00-21.00   sec 48.0 MBytes  402 Mbits/sec 35957
[ 5] 21.00-22.00   sec 45.0 MBytes  378 Mbits/sec 33736
[ 5] 22.00-23.00   sec 46.8 MBytes  392 Mbits/sec 35059
[ 5] 23.00-24.00   sec 45.3 MBytes  380 Mbits/sec 33964
[ 5] 24.00-25.00   sec 46.6 MBytes  391 Mbits/sec 34918
[ 5] 25.00-26.00   sec 45.5 MBytes  382 Mbits/sec 34114
[ 5] 26.00-27.00   sec 42.2 MBytes  355 Mbits/sec 31644
[ 5] 27.00-28.00   sec 38.0 MBytes  318 Mbits/sec 28437
[ 5] 28.00-29.00   sec 44.0 MBytes  369 Mbits/sec 32924
[ 5] 29.00-30.00   sec 44.2 MBytes  371 Mbits/sec 33126
[ 5] 30.00-31.00   sec 46.2 MBytes  388 Mbits/sec 34614
[ 5] 31.00-32.00   sec 44.9 MBytes  376 Mbits/sec 33614
[ 5] 32.00-33.00   sec 45.0 MBytes  377 Mbits/sec 33699
[ 5] 33.00-34.00   sec 44.4 MBytes  372 Mbits/sec 33253
[ 5] 34.00-35.00   sec 45.7 MBytes  383 Mbits/sec 34252
[ 5] 35.00-36.00   sec 47.2 MBytes  396 Mbits/sec 35354
[ 5] 36.00-37.00   sec 46.7 MBytes  391 Mbits/sec 34941
[ 5] 37.00-38.00   sec 43.1 MBytes  362 Mbits/sec 32307
[ 5] 38.00-39.00   sec 46.2 MBytes  387 Mbits/sec 34581
[ 5] 39.00-40.00   sec 45.6 MBytes  382 Mbits/sec 34139

[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-40.00    sec 1.75 GBytes  377 Mbits/sec 0.000 ms    0/1346010 (0%) sender
[ 5] 0.00-40.00    sec 1.75 GBytes  375 Mbits/sec 0.056 ms    5211/1346010 (0.39%) receiver

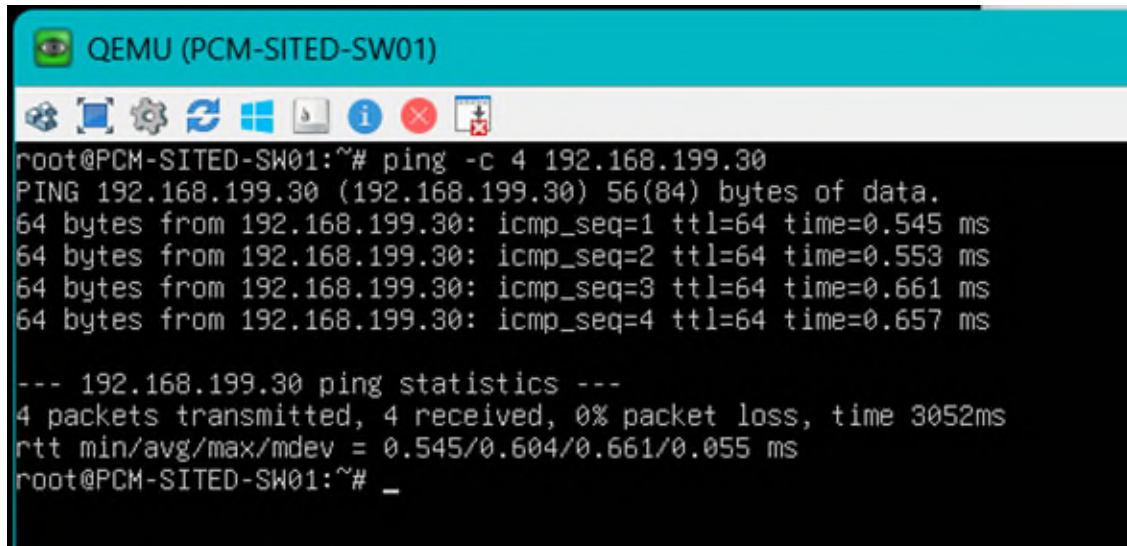
iperf Done.
root@PCM-SITEC-SW01:~#

```

Figure 12 – screenshot of PCM-SITEC-SW01 jitter test

PCM-SITED-SW01

PCM-SITED-SW01 Ping Test



```
QEMU (PCM-SITED-SW01)
root@PCM-SITED-SW01:~# ping -c 4 192.168.199.30
PING 192.168.199.30 (192.168.199.30) 56(84) bytes of data.
64 bytes from 192.168.199.30: icmp_seq=1 ttl=64 time=0.545 ms
64 bytes from 192.168.199.30: icmp_seq=2 ttl=64 time=0.553 ms
64 bytes from 192.168.199.30: icmp_seq=3 ttl=64 time=0.661 ms
64 bytes from 192.168.199.30: icmp_seq=4 ttl=64 time=0.657 ms

--- 192.168.199.30 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.545/0.604/0.661/0.055 ms
root@PCM-SITED-SW01:~# _
```

Figure 13 – screenshot of PCM-SITED-SW01 ping test

PCM-SITED-SW01 Speed Test

QEMU (PCM-SITED-SW01)

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	98.0 MBytes	822 Mb/s	0	757 KBytes
[5]	1.00-2.00	sec	100 MBytes	840 Mb/s	0	757 KBytes
[5]	2.00-3.00	sec	102 MBytes	854 Mb/s	0	795 KBytes
[5]	3.00-4.00	sec	101 MBytes	845 Mb/s	0	836 KBytes
[5]	4.00-5.00	sec	104 MBytes	873 Mb/s	0	836 KBytes
[5]	5.00-6.00	sec	103 MBytes	864 Mb/s	0	836 KBytes
[5]	6.00-7.00	sec	101 MBytes	849 Mb/s	0	836 KBytes
[5]	7.00-8.00	sec	104 MBytes	872 Mb/s	0	836 KBytes
[5]	8.00-9.00	sec	101 MBytes	851 Mb/s	0	836 KBytes
[5]	9.00-10.00	sec	104 MBytes	873 Mb/s	0	836 KBytes
[5]	10.00-11.00	sec	104 MBytes	875 Mb/s	0	836 KBytes
[5]	11.00-12.00	sec	101 MBytes	850 Mb/s	0	1.22 MBytes
[5]	12.00-13.00	sec	104 MBytes	873 Mb/s	0	1.22 MBytes
[5]	13.00-14.00	sec	100 MBytes	839 Mb/s	0	1.22 MBytes
[5]	14.00-15.00	sec	104 MBytes	874 Mb/s	0	1.22 MBytes
[5]	15.00-16.00	sec	104 MBytes	875 Mb/s	0	1.22 MBytes
[5]	16.00-17.00	sec	104 MBytes	875 Mb/s	0	1.22 MBytes
[5]	17.00-18.00	sec	102 MBytes	851 Mb/s	0	1.22 MBytes
[5]	18.00-19.00	sec	103 MBytes	863 Mb/s	0	1.22 MBytes
[5]	19.00-20.00	sec	103 MBytes	862 Mb/s	0	1.22 MBytes
[5]	20.00-21.00	sec	104 MBytes	872 Mb/s	0	1.22 MBytes
[5]	21.00-22.00	sec	104 MBytes	874 Mb/s	0	1.22 MBytes
[5]	22.00-23.00	sec	103 MBytes	863 Mb/s	0	1.22 MBytes
[5]	23.00-24.00	sec	104 MBytes	872 Mb/s	0	1.22 MBytes
[5]	24.00-25.00	sec	104 MBytes	873 Mb/s	0	1.22 MBytes
[5]	25.00-26.00	sec	103 MBytes	864 Mb/s	0	1.22 MBytes
[5]	26.00-27.00	sec	96.1 MBytes	806 Mb/s	0	1.22 MBytes
[5]	27.00-28.00	sec	101 MBytes	844 Mb/s	0	1.22 MBytes
[5]	28.00-29.00	sec	101 MBytes	848 Mb/s	0	1.83 MBytes
[5]	29.00-30.00	sec	103 MBytes	861 Mb/s	0	1.83 MBytes
[5]	30.00-31.00	sec	104 MBytes	874 Mb/s	0	1.83 MBytes
[5]	31.00-32.00	sec	103 MBytes	861 Mb/s	0	2.78 MBytes
[5]	32.00-33.00	sec	102 MBytes	852 Mb/s	0	2.78 MBytes
[5]	33.00-34.00	sec	104 MBytes	875 Mb/s	0	2.78 MBytes
[5]	34.00-35.00	sec	104 MBytes	871 Mb/s	0	2.78 MBytes
[5]	35.00-36.00	sec	104 MBytes	873 Mb/s	0	2.78 MBytes
[5]	36.00-37.00	sec	101 MBytes	846 Mb/s	0	2.78 MBytes
[5]	37.00-38.00	sec	104 MBytes	873 Mb/s	0	2.78 MBytes
[5]	38.00-39.00	sec	103 MBytes	863 Mb/s	0	2.78 MBytes
[5]	39.00-40.00	sec	104 MBytes	872 Mb/s	0	2.78 MBytes
[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	4.01 GBytes	860 Mb/s	0	sender
[5]	0.00-40.01	sec	4.00 GBytes	860 Mb/s	0	receiver

iperf Done.
root@PCM-SITED-SW01:~#

Figure 14 – screenshot of PCM-SITED-SW01 speed test

PCM-SITED-SW01 Jitter Test

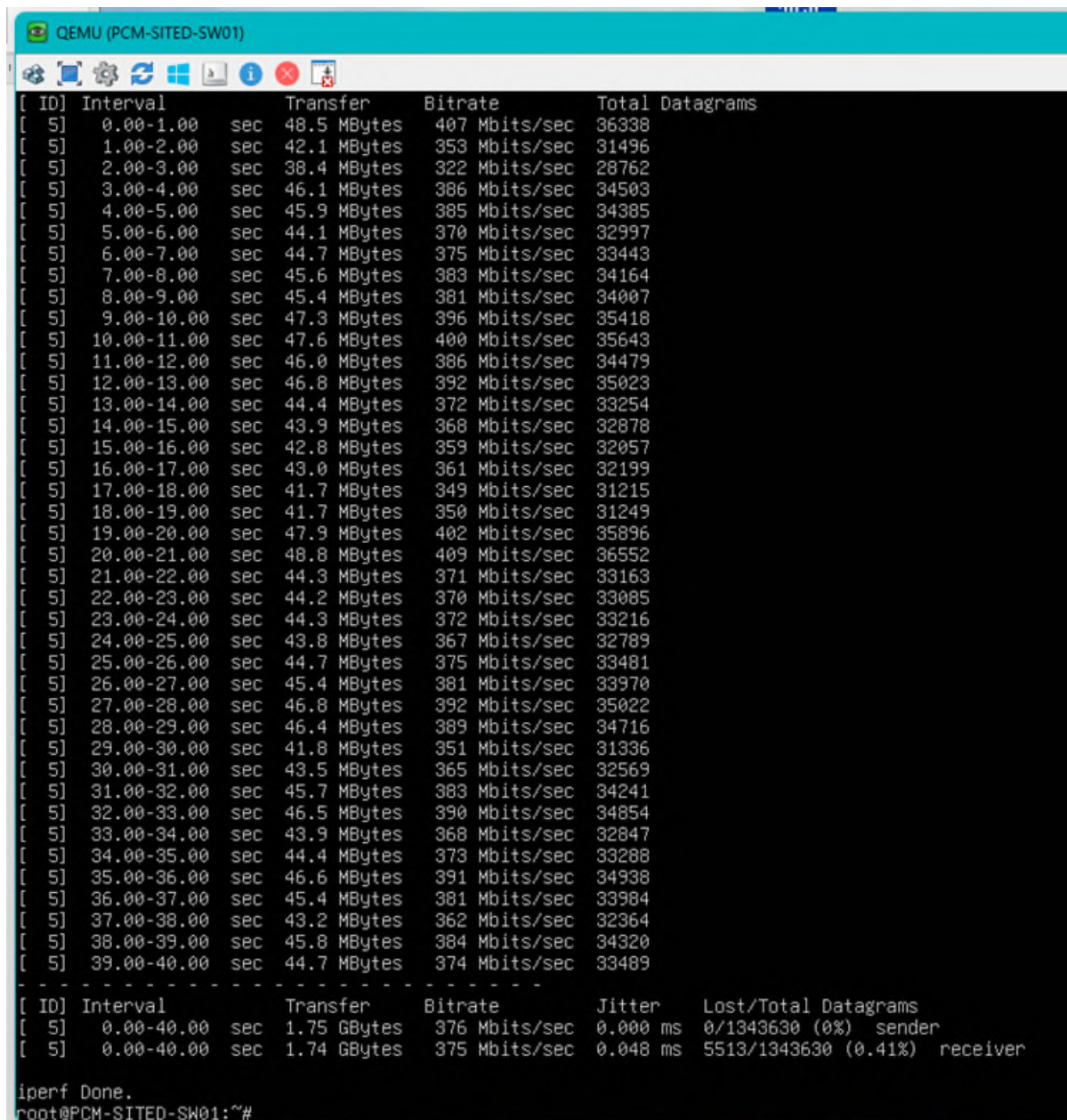
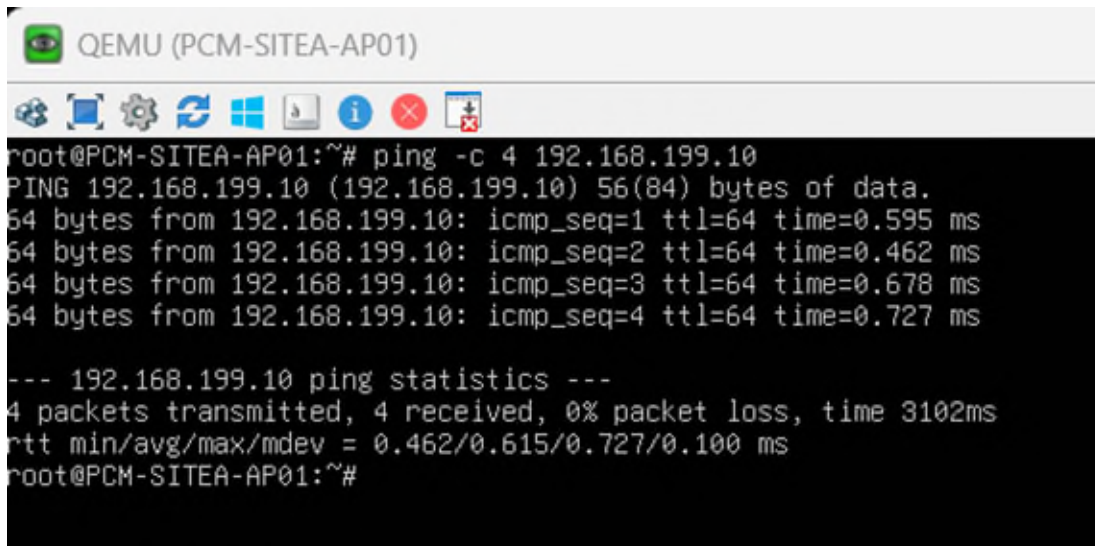


Figure 15 – screenshot of PCM-SITED-SW01 speed test

PCM-SITEA-AP01

PCM-SITEA-AP01 Ping Test



The screenshot shows a terminal window titled "QEMU (PCM-SITEA-AP01)". The terminal output displays a successful ping test. The command executed is "ping -c 4 192.168.199.10". The output shows four successful replies from 192.168.199.10, each 64 bytes in size, with varying round-trip times (0.595 ms, 0.462 ms, 0.678 ms, and 0.727 ms). The statistics summary indicates that 4 packets were transmitted and received with 0% packet loss and a total time of 3102ms. The round-trip time statistics are: min/avg/max/mdev = 0.462/0.615/0.727/0.100 ms.

```
root@PCM-SITEA-AP01:~# ping -c 4 192.168.199.10
PING 192.168.199.10 (192.168.199.10) 56(84) bytes of data:
64 bytes from 192.168.199.10: icmp_seq=1 ttl=64 time=0.595 ms
64 bytes from 192.168.199.10: icmp_seq=2 ttl=64 time=0.462 ms
64 bytes from 192.168.199.10: icmp_seq=3 ttl=64 time=0.678 ms
64 bytes from 192.168.199.10: icmp_seq=4 ttl=64 time=0.727 ms

--- 192.168.199.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3102ms
rtt min/avg/max/mdev = 0.462/0.615/0.727/0.100 ms
root@PCM-SITEA-AP01:~#
```

Figure 16 – screenshot of PCM-SITEA-AP01 ping test

PCM-SITEA-AP02 Speed Test

QEMU (PCM-SITEA-AP01)

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	91.5 MBytes	767 Mb/s	0	7.15 MBytes
[5]	1.00-2.00	sec	104 MBytes	872 Mb/s	0	7.15 MBytes
[5]	2.00-3.00	sec	104 MBytes	875 Mb/s	0	7.15 MBytes
[5]	3.00-4.00	sec	102 MBytes	860 Mb/s	0	7.15 MBytes
[5]	4.00-5.00	sec	104 MBytes	874 Mb/s	0	7.15 MBytes
[5]	5.00-6.00	sec	102 MBytes	859 Mb/s	0	7.15 MBytes
[5]	6.00-7.00	sec	103 MBytes	864 Mb/s	0	7.15 MBytes
[5]	7.00-8.00	sec	100 MBytes	840 Mb/s	0	7.15 MBytes
[5]	8.00-9.00	sec	103 MBytes	866 Mb/s	0	7.15 MBytes
[5]	9.00-10.00	sec	103 MBytes	865 Mb/s	0	7.15 MBytes
[5]	10.00-11.00	sec	96.0 MBytes	806 Mb/s	0	7.15 MBytes
[5]	11.00-12.00	sec	104 MBytes	872 Mb/s	0	7.15 MBytes
[5]	12.00-13.00	sec	100 MBytes	842 Mb/s	0	7.15 MBytes
[5]	13.00-14.00	sec	104 MBytes	873 Mb/s	0	7.15 MBytes
[5]	14.00-15.00	sec	104 MBytes	873 Mb/s	0	7.15 MBytes
[5]	15.00-16.00	sec	104 MBytes	871 Mb/s	0	7.15 MBytes
[5]	16.00-17.00	sec	104 MBytes	875 Mb/s	0	7.15 MBytes
[5]	17.00-18.00	sec	102 MBytes	852 Mb/s	0	7.15 MBytes
[5]	18.00-19.00	sec	104 MBytes	873 Mb/s	0	7.15 MBytes
[5]	19.00-20.00	sec	104 MBytes	872 Mb/s	0	7.15 MBytes
[5]	20.00-21.00	sec	100 MBytes	838 Mb/s	0	7.15 MBytes
[5]	21.00-22.00	sec	104 MBytes	874 Mb/s	0	7.15 MBytes
[5]	22.00-23.00	sec	104 MBytes	871 Mb/s	0	7.15 MBytes
[5]	23.00-24.00	sec	103 MBytes	863 Mb/s	0	7.15 MBytes
[5]	24.00-25.00	sec	104 MBytes	871 Mb/s	0	7.15 MBytes
[5]	25.00-26.00	sec	100 MBytes	841 Mb/s	0	7.15 MBytes
[5]	26.00-27.00	sec	103 MBytes	861 Mb/s	0	7.15 MBytes
[5]	27.00-28.00	sec	101 MBytes	850 Mb/s	0	7.15 MBytes
[5]	28.00-29.00	sec	104 MBytes	872 Mb/s	0	7.15 MBytes
[5]	29.00-30.00	sec	104 MBytes	875 Mb/s	0	7.15 MBytes
[5]	30.00-31.00	sec	103 MBytes	862 Mb/s	0	7.15 MBytes
[5]	31.00-32.00	sec	102 MBytes	860 Mb/s	0	7.15 MBytes
[5]	32.00-33.00	sec	104 MBytes	868 Mb/s	0	7.15 MBytes
[5]	33.00-34.00	sec	104 MBytes	872 Mb/s	0	7.15 MBytes
[5]	34.00-35.00	sec	104 MBytes	874 Mb/s	0	7.15 MBytes
[5]	35.00-36.00	sec	104 MBytes	873 Mb/s	0	7.15 MBytes
[5]	36.00-37.00	sec	104 MBytes	875 Mb/s	0	7.15 MBytes
[5]	37.00-38.00	sec	103 MBytes	867 Mb/s	0	7.15 MBytes
[5]	38.00-39.00	sec	104 MBytes	877 Mb/s	0	7.15 MBytes
[5]	39.00-40.00	sec	103 MBytes	862 Mb/s	0	7.15 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	4.01 GBytes	861 Mb/s	0	sender
[5]	0.00-40.01	sec	4.01 GBytes	861 Mb/s		receiver

iperf Done.
root@PCM-SITEA-AP01:~#

Figure 17 – screenshot of PCM-SITEA-AP01 speed test

PCM-SITEA-AP01 Jitter Test

```

QEMU (PCM-SITEA-AP01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00 sec 46.3 MBytes 388 Mbits/sec 34710
[ 5] 1.00-2.00 sec 46.8 MBytes 392 Mbits/sec 35015
[ 5] 2.00-3.00 sec 46.0 MBytes 386 Mbits/sec 34458
[ 5] 3.00-4.00 sec 46.6 MBytes 391 Mbits/sec 34937
[ 5] 4.00-5.00 sec 46.4 MBytes 389 Mbits/sec 34766
[ 5] 5.00-6.00 sec 43.6 MBytes 366 Mbits/sec 32678
[ 5] 6.00-7.00 sec 47.5 MBytes 399 Mbits/sec 35543
[ 5] 7.00-8.00 sec 47.3 MBytes 397 Mbits/sec 35435
[ 5] 8.00-9.00 sec 45.6 MBytes 382 Mbits/sec 34136
[ 5] 9.00-10.00 sec 47.5 MBytes 399 Mbits/sec 35599
[ 5] 10.00-11.00 sec 47.1 MBytes 396 Mbits/sec 35314
[ 5] 11.00-12.00 sec 42.6 MBytes 358 Mbits/sec 31924
[ 5] 12.00-13.00 sec 44.9 MBytes 376 Mbits/sec 33625
[ 5] 13.00-14.00 sec 44.1 MBytes 371 Mbits/sec 33052
[ 5] 14.00-15.00 sec 45.8 MBytes 384 Mbits/sec 34330
[ 5] 15.00-16.00 sec 46.6 MBytes 391 Mbits/sec 34911
[ 5] 16.00-17.00 sec 44.3 MBytes 372 Mbits/sec 33201
[ 5] 17.00-18.00 sec 44.6 MBytes 374 Mbits/sec 33390
[ 5] 18.00-19.00 sec 46.2 MBytes 388 Mbits/sec 34569
[ 5] 19.00-20.00 sec 47.8 MBytes 401 Mbits/sec 35783
[ 5] 20.00-21.00 sec 45.9 MBytes 385 Mbits/sec 34352
[ 5] 21.00-22.01 sec 44.7 MBytes 371 Mbits/sec 33493
[ 5] 22.01-23.00 sec 46.3 MBytes 393 Mbits/sec 34669
[ 5] 23.00-24.00 sec 41.7 MBytes 349 Mbits/sec 31201
[ 5] 24.00-25.00 sec 43.9 MBytes 369 Mbits/sec 32917
[ 5] 25.00-26.00 sec 44.6 MBytes 374 Mbits/sec 33392
[ 5] 26.00-27.00 sec 44.1 MBytes 370 Mbits/sec 32993
[ 5] 27.00-28.00 sec 45.3 MBytes 380 Mbits/sec 33921
[ 5] 28.00-29.00 sec 42.4 MBytes 355 Mbits/sec 31721
[ 5] 29.00-30.00 sec 42.9 MBytes 360 Mbits/sec 32137
[ 5] 30.00-31.00 sec 44.2 MBytes 371 Mbits/sec 33140
[ 5] 31.00-32.00 sec 40.9 MBytes 343 Mbits/sec 30657
[ 5] 32.00-33.00 sec 44.6 MBytes 374 Mbits/sec 33377
[ 5] 33.00-34.00 sec 45.9 MBytes 385 Mbits/sec 34376
[ 5] 34.00-35.00 sec 42.4 MBytes 356 Mbits/sec 31770
[ 5] 35.00-36.00 sec 45.2 MBytes 379 Mbits/sec 33867
[ 5] 36.00-37.00 sec 41.6 MBytes 349 Mbits/sec 31150
[ 5] 37.00-38.00 sec 46.2 MBytes 388 Mbits/sec 34610
[ 5] 38.00-39.00 sec 47.2 MBytes 396 Mbits/sec 35380
[ 5] 39.00-40.00 sec 44.6 MBytes 374 Mbits/sec 33439
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-40.00 sec 1.76 GBytes 378 Mbits/sec 0.000 ms 0/1349938 (0%) sender
[ 5] 0.00-40.00 sec 1.75 GBytes 377 Mbits/sec 0.045 ms 5051/1349938 (0.37%) receiver

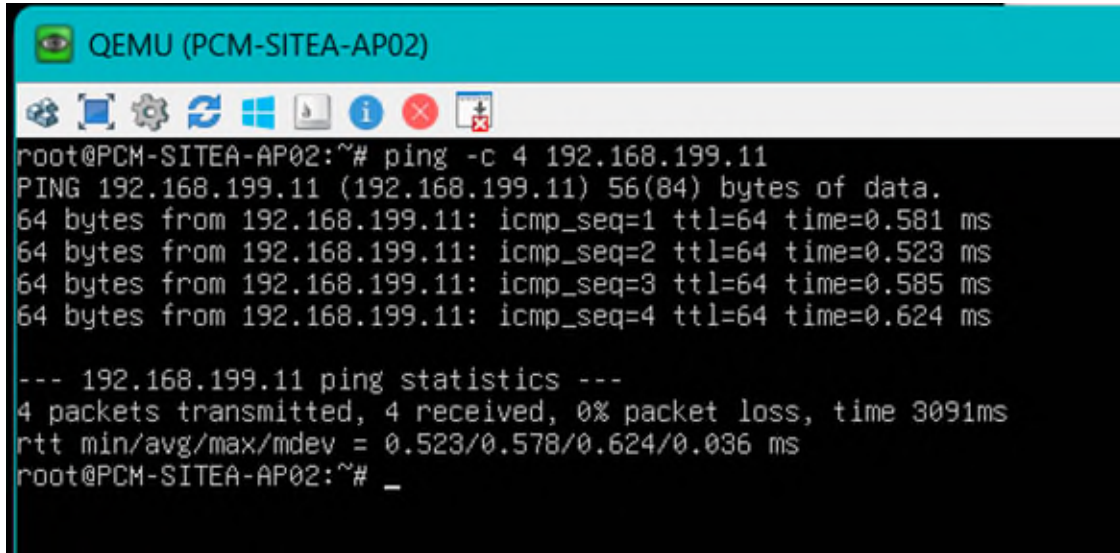
iperf Done.
root@PCM-SITEA-AP01:~#

```

Figure 18 – screenshot of PCM-SITEA-AP01 jitter test

PCM-SITEA-AP02

PCM-SITEA-AP02 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITEA-AP02)". The terminal displays the output of a ping command: `ping -c 4 192.168.199.11`. The output shows four successful ping requests, each with a 64-byte payload, an ICMP sequence number, a TTL of 64, and a response time. The statistics section indicates that all four packets were received with 0% packet loss and a total time of 3091ms. The round-trip times (RTT) are listed as min/avg/max/mdev = 0.523/0.578/0.624/0.036 ms.

```
root@PCM-SITEA-AP02:~# ping -c 4 192.168.199.11
PING 192.168.199.11 (192.168.199.11) 56(84) bytes of data.
64 bytes from 192.168.199.11: icmp_seq=1 ttl=64 time=0.581 ms
64 bytes from 192.168.199.11: icmp_seq=2 ttl=64 time=0.523 ms
64 bytes from 192.168.199.11: icmp_seq=3 ttl=64 time=0.585 ms
64 bytes from 192.168.199.11: icmp_seq=4 ttl=64 time=0.624 ms

--- 192.168.199.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3091ms
rtt min/avg/max/mdev = 0.523/0.578/0.624/0.036 ms
root@PCM-SITEA-AP02:~# _
```

Figure 19 – screenshot of PCM-SITEA-AP02 ping test

PCM-SITEA-AP02 Speed Test

```

QEMU (PCM-SITEA-AP02)

[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 5]  0.00-1.00    sec    103 MBytes    861 Mbits/sec    0    426 KBytes
[ 5]  1.00-2.00    sec    105 MBytes    881 Mbits/sec    0    675 KBytes
[ 5]  2.00-3.00    sec    104 MBytes    868 Mbits/sec    0    744 KBytes
[ 5]  3.00-4.00    sec    103 MBytes    861 Mbits/sec    0    744 KBytes
[ 5]  4.00-5.00    sec    104 MBytes    871 Mbits/sec    0    744 KBytes
[ 5]  5.00-6.00    sec   98.8 MBytes    829 Mbits/sec    0    744 KBytes
[ 5]  6.00-7.00    sec    104 MBytes    871 Mbits/sec    0    744 KBytes
[ 5]  7.00-8.00    sec    103 MBytes    862 Mbits/sec    0    744 KBytes
[ 5]  8.00-9.00    sec    103 MBytes    867 Mbits/sec    0    782 KBytes
[ 5]  9.00-10.00   sec    103 MBytes    862 Mbits/sec    0    782 KBytes
[ 5] 10.00-11.00   sec    104 MBytes    873 Mbits/sec    0    782 KBytes
[ 5] 11.00-12.00   sec    103 MBytes    862 Mbits/sec    0    782 KBytes
[ 5] 12.00-13.00   sec    104 MBytes    873 Mbits/sec    0    782 KBytes
[ 5] 13.00-14.00   sec    103 MBytes    863 Mbits/sec    0    782 KBytes
[ 5] 14.00-15.00   sec    104 MBytes    875 Mbits/sec    0    782 KBytes
[ 5] 15.00-16.00   sec   88.0 MBytes    738 Mbits/sec    0    1.14 MBytes
[ 5] 16.00-17.00   sec    103 MBytes    863 Mbits/sec    0    1.14 MBytes
[ 5] 17.00-18.00   sec    103 MBytes    862 Mbits/sec    0    1.14 MBytes
[ 5] 18.00-19.00   sec    103 MBytes    863 Mbits/sec    0    1.14 MBytes
[ 5] 19.00-20.00   sec   99.0 MBytes    830 Mbits/sec    0    1.14 MBytes
[ 5] 20.00-21.00   sec   99.1 MBytes    832 Mbits/sec    0    1.14 MBytes
[ 5] 21.00-22.00   sec    103 MBytes    863 Mbits/sec    0    1.14 MBytes
[ 5] 22.00-23.00   sec    103 MBytes    867 Mbits/sec    0    1.14 MBytes
[ 5] 23.00-24.00   sec    102 MBytes    852 Mbits/sec    0    1.14 MBytes
[ 5] 24.00-25.00   sec    103 MBytes    862 Mbits/sec    0    1.14 MBytes
[ 5] 25.00-26.00   sec   99.0 MBytes    830 Mbits/sec    0    1.14 MBytes
[ 5] 26.00-27.00   sec    103 MBytes    863 Mbits/sec    0    1.14 MBytes
[ 5] 27.00-28.00   sec    103 MBytes    865 Mbits/sec    0    1.14 MBytes
[ 5] 28.00-29.00   sec    103 MBytes    864 Mbits/sec    0    1.14 MBytes
[ 5] 29.00-30.00   sec    104 MBytes    876 Mbits/sec    0    1.14 MBytes
[ 5] 30.00-31.00   sec   97.9 MBytes    821 Mbits/sec    0    1.14 MBytes
[ 5] 31.00-32.00   sec    104 MBytes    874 Mbits/sec    0    1.14 MBytes
[ 5] 32.00-33.00   sec    103 MBytes    862 Mbits/sec    0    1.14 MBytes
[ 5] 33.00-34.00   sec    103 MBytes    863 Mbits/sec    0    1.71 MBytes
[ 5] 34.00-35.00   sec    103 MBytes    864 Mbits/sec    0    1.71 MBytes
[ 5] 35.00-36.00   sec    103 MBytes    862 Mbits/sec    0    1.71 MBytes
[ 5] 36.00-37.00   sec    103 MBytes    862 Mbits/sec    0    1.71 MBytes
[ 5] 37.00-38.00   sec   98.1 MBytes    823 Mbits/sec    0    1.71 MBytes
[ 5] 38.00-39.00   sec    102 MBytes    852 Mbits/sec    0    1.71 MBytes
[ 5] 39.00-40.00   sec    102 MBytes    853 Mbits/sec    0    1.71 MBytes

[ ID] Interval      Transfer      Bitrate      Retr
[ 5]  0.00-40.00   sec    3.99 GBytes    856 Mbits/sec    0
[ 5]  0.00-40.01   sec    3.98 GBytes    856 Mbits/sec    0

iperf Done.
root@PCM-SITEA-AP02:~#

```

Figure 20 – screenshot of PCM-SITEA-AP02 speed test

PCM-SITEA-AP02 Jitter Test

```

QEMU (PCM-SITEA-AP02)
[ 5] ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00 sec  44.4 MBytes  372 Mbits/sec 33240
[ 5] 1.00-2.00 sec  42.9 MBytes  360 Mbits/sec 32143
[ 5] 2.00-3.00 sec  45.0 MBytes  377 Mbits/sec 33694
[ 5] 3.00-4.00 sec  46.0 MBytes  386 Mbits/sec 34458
[ 5] 4.00-5.01 sec  44.5 MBytes  370 Mbits/sec 33319
[ 5] 5.01-6.00 sec  47.0 MBytes  397 Mbits/sec 35168
[ 5] 6.00-7.00 sec  47.5 MBytes  399 Mbits/sec 35593
[ 5] 7.00-8.00 sec  46.7 MBytes  391 Mbits/sec 34955
[ 5] 8.00-9.00 sec  47.2 MBytes  396 Mbits/sec 35367
[ 5] 9.00-10.00 sec 45.1 MBytes  379 Mbits/sec 33791
[ 5] 10.00-11.00 sec 43.7 MBytes  367 Mbits/sec 32733
[ 5] 11.00-12.00 sec 41.7 MBytes  350 Mbits/sec 31256
[ 5] 12.00-13.00 sec 43.5 MBytes  365 Mbits/sec 32599
[ 5] 13.00-14.00 sec 42.9 MBytes  360 Mbits/sec 32160
[ 5] 14.00-15.00 sec 45.8 MBytes  384 Mbits/sec 34276
[ 5] 15.00-16.00 sec 43.2 MBytes  363 Mbits/sec 32390
[ 5] 16.00-17.00 sec 43.7 MBytes  367 Mbits/sec 32765
[ 5] 17.00-18.00 sec 47.9 MBytes  401 Mbits/sec 35860
[ 5] 18.00-19.00 sec 47.6 MBytes  400 Mbits/sec 35657
[ 5] 19.00-20.00 sec 47.8 MBytes  401 Mbits/sec 35804
[ 5] 20.00-21.00 sec 45.2 MBytes  379 Mbits/sec 33841
[ 5] 21.00-22.00 sec 45.9 MBytes  385 Mbits/sec 34361
[ 5] 22.00-23.00 sec 43.5 MBytes  365 Mbits/sec 32575
[ 5] 23.00-24.00 sec 42.1 MBytes  353 Mbits/sec 31537
[ 5] 24.00-25.00 sec 45.6 MBytes  382 Mbits/sec 34124
[ 5] 25.00-26.00 sec 43.8 MBytes  367 Mbits/sec 32783
[ 5] 26.00-27.00 sec 44.0 MBytes  369 Mbits/sec 32963
[ 5] 27.00-28.00 sec 45.3 MBytes  380 Mbits/sec 33930
[ 5] 28.00-29.00 sec 45.0 MBytes  378 Mbits/sec 33741
[ 5] 29.00-30.00 sec 43.4 MBytes  364 Mbits/sec 32518
[ 5] 30.00-31.00 sec 44.8 MBytes  376 Mbits/sec 33540
[ 5] 31.00-32.00 sec 44.5 MBytes  374 Mbits/sec 33360
[ 5] 32.00-33.00 sec 44.4 MBytes  373 Mbits/sec 33279
[ 5] 33.00-34.00 sec 45.5 MBytes  382 Mbits/sec 34112
[ 5] 34.00-35.00 sec 45.9 MBytes  386 Mbits/sec 34398
[ 5] 35.00-36.00 sec 45.4 MBytes  381 Mbits/sec 34026
[ 5] 36.00-37.00 sec 40.6 MBytes  341 Mbits/sec 30415
[ 5] 37.00-38.00 sec 43.5 MBytes  365 Mbits/sec 32600
[ 5] 38.00-39.00 sec 45.3 MBytes  380 Mbits/sec 33942
[ 5] 39.00-40.00 sec 44.3 MBytes  372 Mbits/sec 33197
-----
[ 5] ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-40.00 sec  1.75 GBytes  376 Mbits/sec 0.000 ms  0/1342470 (0%) sender
[ 5] 0.00-40.00 sec  1.75 GBytes  375 Mbits/sec 0.045 ms 3681/1342470 (0.27%) receiver

iperf Done.
root@PCM-SITEA-AP02:~#

```

Figure 21 – screenshot of PCM-SITEA-AP02 jitter test